STAT 7900: Special Topic: Causal Analysis



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STAT 7900

Recap: the causal question

Descriptive: how do outcomes differ across observed groups?

Predictive: who is likely to have the outcome?

Causal: what would happen if we assigned an intervention?

Specify population, treatment, comparator, outcome, and time horizon.

Example: does Drug A lower four-week blood pressure in eligible adults?

Potential outcomes and the target effect

$Y_i(1)$ and $Y_i(0)$ describe the same person under two treatments.

We observe only

$$Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0).$$

$$ATE = E[Y(1) - Y(0)]$$

in the target population.

$$ATT = E[Y(1) - Y(0) \mid T = 1]$$

among treated people.

$E[Y \mid T = 1] - E[Y \mid T = 0]$ can differ from the ATE.

Four assumptions for causal interpretation

Consistency: treatment received matches the intervention defining $Y(t)$.

Exchangeability:

$$\{Y(1), Y(0)\} \perp T \mid X.$$

Positivity: both treatments occur within relevant X profiles.

No interference: another person's treatment does not change my outcome.

Measured covariate balance alone cannot verify all four assumptions.

Adjustment depends on the causal graph

Role	Example path	For the total effect
Confounder	$T \leftarrow C \rightarrow Y$	Block the backdoor path
Mediator	$T \rightarrow M \rightarrow Y$	Preserve this causal pathway
Collider	$T \rightarrow M \leftarrow U \rightarrow Y$	Avoid opening the closed path

A variable's role depends on the path and the estimand.

Identification through the g-formula

$$E[Y(t)] = E_{\mathrm{x}}\{E[Y \mid T = t, X]\}.$$

Within each X stratum, use comparable observed units to estimate the treatment-specific mean.

Average those means over one common target distribution of X.

The DAG and assumptions justify this replacement of missing counterfactual means.

Identification gives an observed-data formula. Estimation computes it.

g-computation and the treatment coefficient

Fit $m(t, x) = E[Y \mid T = t, X = x]$.

Predict $\hat{m}(1, X_i)$ and $\hat{m}(0, X_i)$ for each person.

Average: $\hat{\tau}_g = n^{-1} \sum_i \{\hat{m}(1, X_i) - \hat{m}(0, X_i)\}$.

With $T \times X$ interaction, the contrast varies with X.

For $m(t, x) = \beta_0 + \beta_T t + \beta_X x + \beta_{TX} tx$, $ATE = \beta_T + \beta_{TX} E[X]$.

Structural causal models

$C := f_C(U_C)$ generates baseline severity.

$T := f_T(C, U_T)$ generates treatment assignment.

$Y := f_Y(C, T, U_Y)$ generates the outcome.

The structural rules state how the system responds to interventions.

A DAG gives the causal structure. An SCM also specifies the mechanisms.

Conditioning, intervention, counterfactual

Operation	Meaning	What stays fixed?
Condition on $T = 1$	Select people who received treatment	Their observed characteristics
$do(T = 1)$	Replace the treatment rule with $T := 1$	Other structural rules
$Y_i(1)$	Outcome for person i under treatment 1	That person's background factors

Intervening changes assignment. Conditioning selects an observed subgroup.

Simulation: a known causal world

C is baseline severity, with

$$C \sim N(0,1).$$

$$P(T = 1 \mid C) = logistic(1.2C).$$

$$Y = 120 + 10C - 6T + \epsilon$$

with independent $\epsilon \sim N(0,25)$.

Higher severity increases treatment probability and blood pressure.

True effect: $Y_i(1) - Y_i(0) = -6\ mmHg$ for every person.

Simulation: generate and compare

```
set.seed(7900)
n <- 5000
C <- rnorm(n)
T <- rbinom(n, 1, plogis(1.2 * C))
eps <- rnorm(n, sd = 5)
Y <- 120 + 10 * C - 6 * T + eps
dat <- data.frame(C, T, Y)
coef(lm(Y ~ T, data = dat))["T"]
coef(lm(Y ~ T + C, data = dat))["T"]
```

Predict the crude and adjusted results before running the code.

Simulation: association versus intervention

Comparison	Approximate result	Reason
Crude observed contrast	$+3.5\ mmHg$	Treated patients have greater severity
Adjustment for C	$-6.0\ mmHg$	Correct outcome model blocks confounding
Same-person interventions	Exactly $-6\ mmHg$	The structural contrast is constant

$$Y_1 = 120 + 10C - 6 + \epsilon \text{ and } Y_0 = 120 + 10C + \epsilon.$$

Randomization and Identification

Under simple random assignment,

$$T \perp \{Y(1), Y(0), X\}.$$

With consistency,

$$E[Y(t)] = E[Y \mid T = t].$$

Therefore

$$ATE = E[Y \mid T = 1] - E[Y \mid T = 0].$$

Randomization balances baseline causes in expectation, not exactly in every sample.

Randomization identifies the assignment effect when outcomes and design support it.

Randomized designs and analysis

Design	Assignment mechanism	Analysis implication
Complete / Bernoulli	Individuals receive random assignments	Respect fixed versus random arm sizes
Blocked / paired	Randomize within strata or pairs	Account for strata or pairs
Cluster	Randomize clinics or other groups	Account for within-cluster dependence

The randomization unit determines how we quantify uncertainty.

Estimation and uncertainty in an RCT

Difference in means: $\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$.

Independent-arm SE: $\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}\right)}$

Example: $\hat{\tau} = -5.3$ and $SE = 1.8\ mmHg$.

Approximate 95% CI: $-5.3 \pm 1.96 \times 1.8 = [-8.83, -1.77]$.

A confidence interval measures sampling uncertainty, not the validity of all assumptions.

Assignment, treatment received, and outcome

Z denotes randomized assignment.
T denotes treatment received.

An intention-to-treat comparison preserves the randomized assignment contrast.

Comparing recipients can reintroduce selection when adherence differs.

A causal effect of treatment received needs additional assumptions or design.

Randomizing Z does not automatically make T unconfounded.

Real Data: Job trainings and earnings

LaLonde data: 614 people, 185 treated and 429 comparison units.

Treatment: participation in the job-training program.

Outcome: earnings in 1978, measured in dollars.

Baseline X includes demographics and earnings in 1974 and 1975.

Question: what would earnings be under training versus no training?

Real data: baseline differences

Variable	Treated	Comparison
Age	25.8	28.0
Married	18.9%	51.3%
Black	84.3%	20.3%
1974 earnings	$2,096	$5,619

1978 earnings: $6,349 versus $6,984. Crude difference ≈ −$635.

Real data: standardize model predictions

```
library(MatchIt)
data("lalonde", package = "MatchIt")
d <- lalonde
fit <- lm(re78 ~ treat + age + educ + race + married +
          nodegree + re74 + re75, data = d)
y1 <- predict(fit, transform(d, treat = 1))
y0 <- predict(fit, transform(d, treat = 0))
mean(y1 - y0)
coef(fit)["treat"]
```

These coincide for this additive linear model without treatment interactions.

Real data: what adjustment establishes

Earlier analysis changed a negative crude contrast to a positive adjusted estimate.

Interpretation still depends on no unmeasured confounding given baseline X.

Poor overlap can make predictions rely heavily on extrapolation.

Model estimates and causal credibility must be assessed separately.

The observed data contain one treatment outcome per person, even after fitting a model.

Recap questions and the next step

Why did the simulated crude estimate have the wrong sign?

When does a treatment coefficient equal the g-computation ATE?

Why can treatment received be confounded in a randomized trial?

What checks would make the job-training comparison more credible?

Next: matching and weighting to create a more comparable observational analysis.

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Propensity scores: the job-training question

Does job training increase earnings in 1978?

$T = 1$ indicates training. $T = 0$ indicates the comparison group.

Baseline X includes demographics and earlier earnings.

Our example uses the 614-person LaLonde dataset throughout.

We need comparable people to stand in for missing counterfactual outcomes.

Why the observed comparison is difficult

Baseline feature	Treated	Comparison
Married	18.9%	51.3%
Black	84.3%	20.3%
1974 earnings	$2,096	$5,619

The groups differ before training begins.

The high-dimensional comparison problem

We want to compare people with similar age, education, and other baseline variables.

Exact agreement on every covariate becomes difficult as X grows.

Earlier earnings are continuous, so exact matches may be rare.

A scalar treatment-assignment summary can help organize comparisons.

The propensity score summarizes assignment using the chosen covariates.

Definition of the propensity score

$$e(x) = P(T = 1 \mid X = x).$$

It is the probability of receiving training given baseline characteristics.

Every participant has a score, whether treated or untreated.

For a given x, the score concerns people sharing that baseline profile.

The observed treatment is 0 or 1. Its probability can lie between them.

Interpreting a score

A participant's estimated score is $\hat{e}(X) = 0.65$.

The fitted model predicts about a 65% training probability for that profile.

The participant can still appear in either treatment arm.

The score does not describe the probability of high earnings or treatment benefit.

A probability of treatment and a causal effect answer different questions.

The score depends on the study sample

The combined sample has 185 treated and 429 comparison participants.

The estimated assignment model describes treatment membership in these study data.

Changing comparison sampling can change the fitted scores.

It does not automatically describe participation probabilities in the whole population.

Specify the study population when interpreting a score.

Which variables belong in the model?

Use baseline covariates justified by the causal question and adjustment graph.

Include measured common causes of training participation and earnings.

Earlier earnings and demographic circumstances motivate the current specification.

Avoid using 1978 earnings, which is the outcome, to predict assignment.

A variable's causal role matters more than its statistical significance.

A logistic treatment model

$$\log \frac{e(X)}{1 - e(X)} = \beta_0 + \beta^{\mathrm{T}} X.$$

The model uses baseline covariates to predict binary treatment membership.

$e(X) = \exp(\eta)/\{1 + \exp(\eta)\}$, where $\eta = \beta_0 + \beta^{\mathrm{T}} X$.

The logistic transform keeps predicted probabilities between 0 and 1.

This regression models treatment assignment, not the outcome.

R: estimate scores in the real dataset

```
library(MatchIt)
data("lalonde", package = "MatchIt")
d <- lalonde
ps_fit <- glm(treat ~ age + educ + race + married +
              nodegree + re74 + re75,
              family = binomial(), data = d)
d$ps <- predict(ps_fit, type = "response")
head(d[c("treat", "age", "educ", "ps")])
```

type = "response" returns probabilities, not log odds.

Fitted scores

Participant	Training	Age	Education	ê(X)
NSW1	1	37	11	0.639
NSW11	1	19	9	0.689
PSID1	0	30	12	0.026
PSID16	0	42	0	0.056

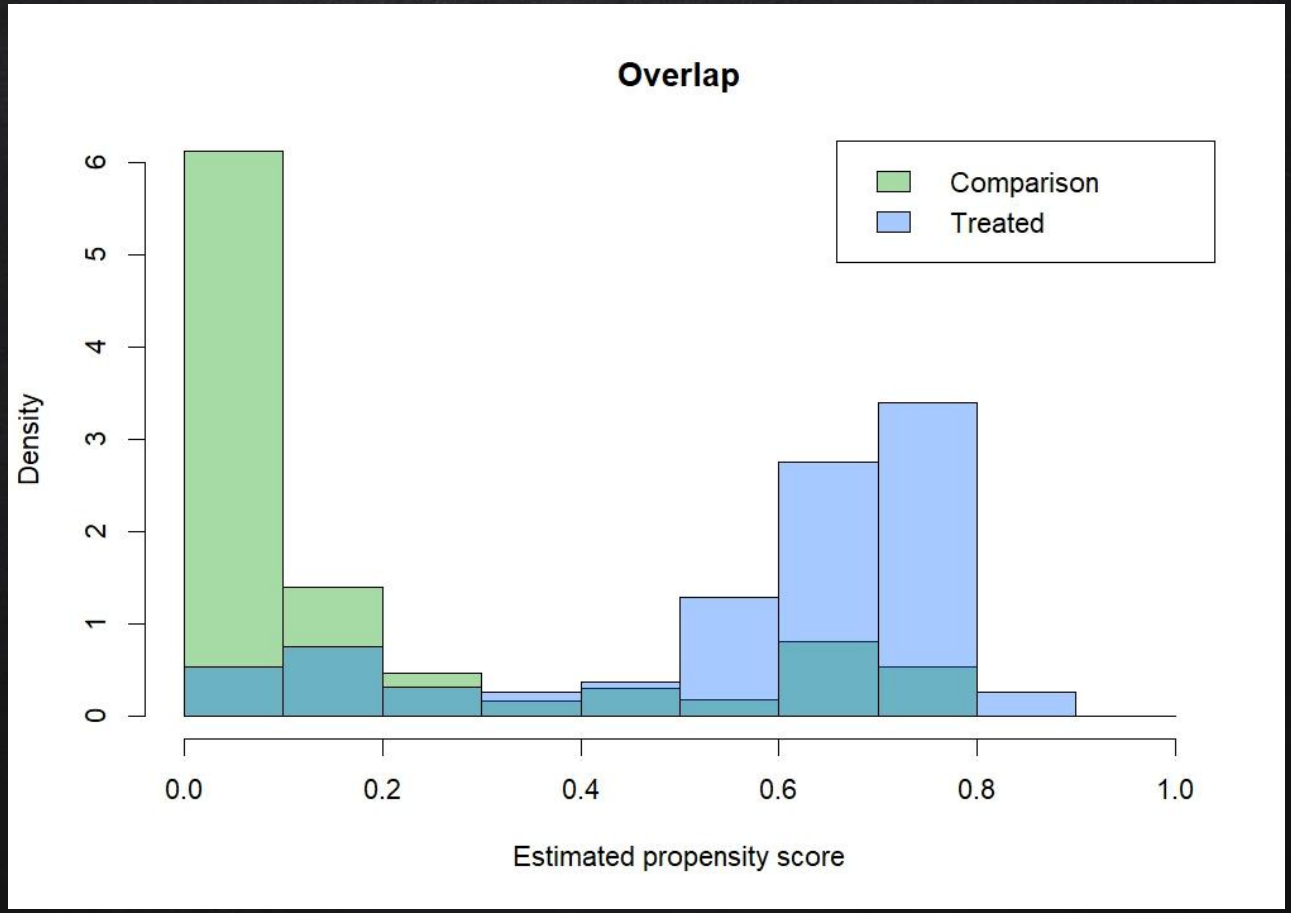
All listed scores also use race, marriage, degree status, and earlier earnings.

Scores differ across observed groups

Group	n	Mean score	Median score	Observed score range
Treated	185	0.577	0.654	0.025–0.853
Comparison	429	0.182	0.076	0.009–0.789

Estimating a propensity score does not itself change either group.

Overlap in the LaLonde data



Positivity and empirical overlap

Population positivity: $0 < e(X) < 1$ for relevant profiles.

Empirical overlap: the sample contains comparable units from both arms.

A logistic model can return $\hat{e} = 0.10$ even for a sparsely supported profile.

Finite scores between 0 and 1 do not establish adequate data support.

A fitted probability cannot create missing counterfactual information.

The propensity score’s balancing property

For the true propensity score:

$$T \perp X \mid e(X).$$

Within a score stratum, treatment assignment does not further select the distribution of X.

This is a population statement, not exact equality between individual profiles.

Estimated scores and finite samples require diagnostic checks.

Comparable score strata can organize a fairer comparison.

Why the balancing property holds

Once $e(X) = s$, every profile in that stratum has treatment probability s.

$$P(T = 1 \mid X, e(X) = s) = s.$$

$$P(T = 1 \mid e(X) = s) = s.$$

Knowing X within the stratum supplies no further assignment information.

Therefore $T \perp X \mid e(X)$.

From balance to causal identification

Assume $\{Y(1), Y(0)\} \perp T \mid X$.

Together with the true balancing score, exchangeability also holds given e(X).

Then $E[Y(t)] = E_{\mathrm{e}}\{E[Y \mid T = t, e(X)]\}$.

Consistency and positivity remain necessary.

The score reduces the adjustment dimension under the causal assumptions.

Similar scores do not mean identical people

Different covariate profiles can produce similar treatment probabilities.

Matching on a score can leave important covariates imbalanced in a sample.

Assess age, prior earnings, and other adjustment covariates individually.

An unmeasured confounder cannot enter this score unless we measure it.

Score similarity is useful, but observed balance still needs checking.

Propensity-score checkpoint

What does a score of 0.65 mean for an untreated participant?

Why do the treated and comparison arms still have different mean scores?

Does a score strictly between 0 and 1 prove positivity?

What would you inspect before interpreting a matched comparison?

Estimating scores is the first step. Comparable groups require a design and balance checks.

Matching: a comparable job-training group

We have estimated each person's probability of receiving training.

Now select untreated participants with similar baseline profiles.

Use their earnings to inform the missing no-training comparison.

The scientific goal remains an effect in a clearly defined population.

Matching uses observed comparison people to support the counterfactual contrast.

Why match when we have g-computation?

Approach	How it constructs the comparison
G-computation	Predict both outcomes using an outcome model
Matching	Select comparable observed treatment and comparison units
Matching + regression	Model outcomes after constructing comparable groups

Matching is an alternative adjustment strategy, not a prerequisite for g-computation.

Benefits and costs of matching

Closer observed comparisons can reduce reliance on outcome extrapolation.

Balance diagnostics make the comparison design visible.

Discarding participants can reduce precision and change the target population.

Matching choices and residual covariate imbalance can affect estimates.

Matching trades sample retention against comparability.

The causal target: ATT

$$ATT = E[Y(1) - Y(0) \mid T = 1].$$

Use treated participants as the focal group.

Find controls to represent their outcomes without training.

Dropping treated participants can change the effect being targeted.

Full ATT and the effect among retained treated participants may differ.

Nearest-neighbor propensity-score matching

For treated participant i, seek a control j with a nearby score.

Distance:

$$d(i,j) = \left|\hat{e}(X_i) - \hat{e}(X_j)\right|.$$

One-to-one matching assigns one eligible control to each retained treated unit.

A greedy algorithm selects each match without optimizing the whole set.

A nearby score is a starting point. Covariate balance determines usefulness.

Actual pairs from the job-training data

Treated ID	Score	Comparison ID	Score	Gap
NSW178	0.853	PSID118	0.789	0.064
NSW95	0.819	PSID226	0.779	0.041
NSW176	0.816	PSID196	0.764	0.052

These are the first three pairs when matching treated scores from largest to smallest.

Matching order and competition for controls

Without replacement, each control can serve only one treated unit.

Early matches remove controls from the pool for later matches.

Different matching orders can produce different pairs.

Record the order so the design is reproducible.

Our example uses descending treated scores: m.order = "largest".

With or without replacement

Choice	Benefit	Cost
Without replacement	Each control appears in one pair	Later treated units may receive weaker matches
With replacement	Good controls can serve multiple treated units	Reuse reduces independent information

A reused control remains one person, even if they appear in several matches.

Calipers: a maximum acceptable distance

A raw score caliper of 0.05 requires $|\hat{e}_i - \hat{e}_j| \leq 0.05$.

A treated person with no eligible control remains unmatched.

Smaller calipers can prevent weak matches but discard more participants.

A caliper can instead be defined on log odds or in SD units.

The number 0.05 has meaning only after specifying the distance scale.

R: the initial matching design

```
library(MatchIt)
data("lalonde", package = "MatchIt")
d <- lalonde
f <- treat ~ age + educ + race + married +
             nodegree + re74 + re75
m0 <- matchit(f, data = d, method = "nearest",
              distance = "glm", estimand = "ATT",
              ratio = 1, replace = FALSE,
              m.order = "largest")
summary(m0)
```

Outcome earnings do not enter the matching formula.

Who remains after initial matching?

Group	Before	After
Treated	185	185
Comparison	429	185
Unused comparison participants	0	244

All treated participants remain, but retention does not establish adequate balance.

Covariate balance: standardized differences

SMD =
(treated mean − comparison mean)/sref.

Use a fixed reference scale when comparing designs.

Here sref is the SD in the original treated group.

Inspect each adjustment covariate. $|SMD| < 0.10$ is a guideline.

A small score difference does not guarantee small covariate differences.

Initial matching improves some covariates

Covariate	Before |SMD|	After |SMD|
Age	0.309	0.072
Married	0.824	0.055
1974 earnings	0.721	0.050
Black indicator	1.757	1.023

Substantial imbalance remains. We should revise the design before relying on its effect estimate.

R: add a raw-score caliper

```
m1 <- matchit(f, data = d, method = "nearest",
              distance = "glm", estimand = "ATT",
              ratio = 1, replace = FALSE,
              m.order = "largest",
              caliper = 0.05, std.caliper = FALSE)
summary(m1)
plot(summary(m1), abs = TRUE)
```

This caliper limits probability differences to 0.05.

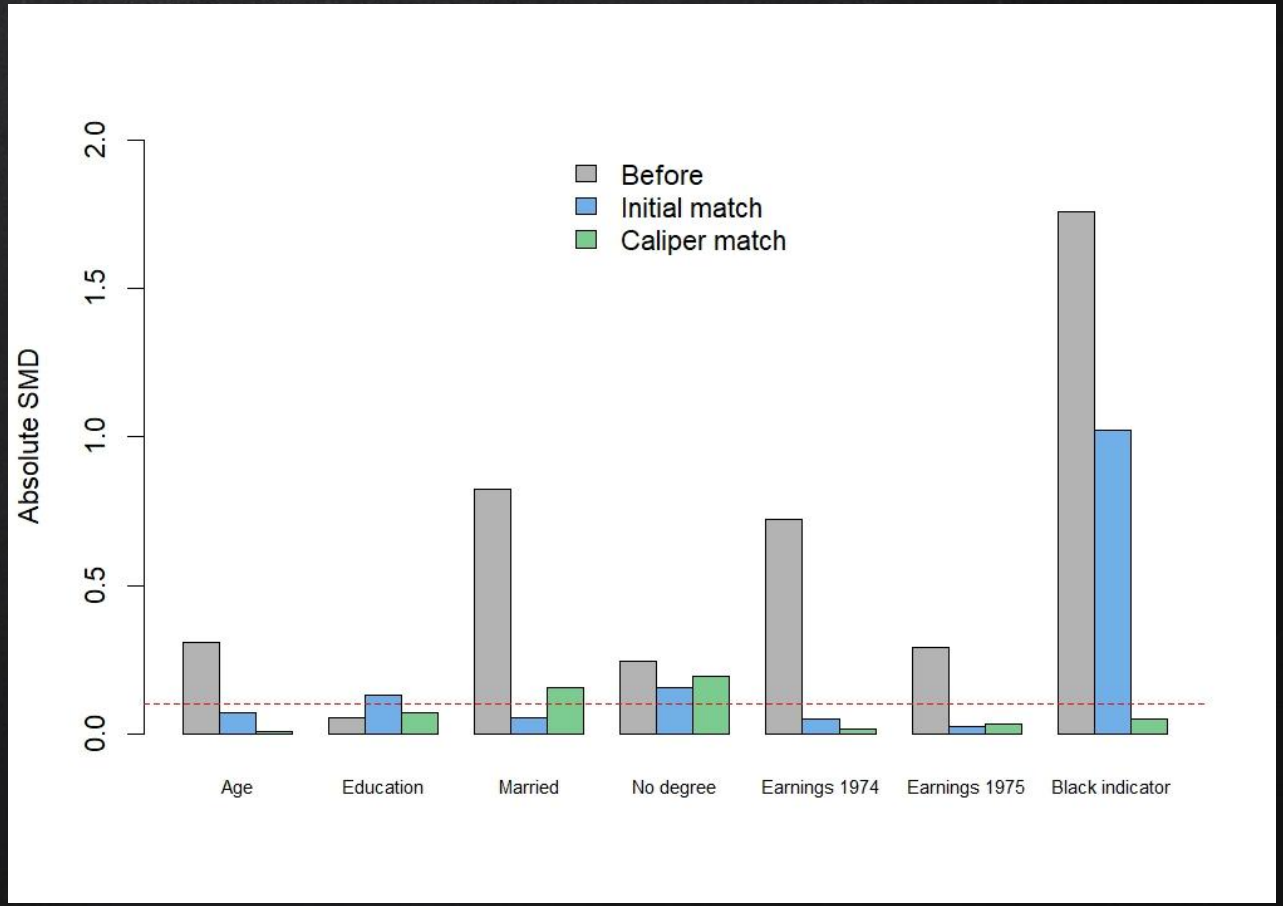
Caliper matching changes the analysis sample

Group	Initial match	0.05 caliper
Treated retained	185	113
Comparison retained	185	113
Treated unmatched	0	72

The caliper design represents retained treated participants, not automatically all 185.

Balance across matching designs

The caliper improves some imbalances, but degree status and marriage still need attention.



If balance is still poor

Revise nonlinear terms or interactions in the propensity model.

Consider exact matching on scientifically important variables.

Consider replacement, a different distance, or a different matching method.

Reassess retention and the target population after each design change.

Improve the comparison using diagnostics, not the size or sign of the estimated effect.

The matched outcome contrast

For each retained pair, compute $D_i = Y_{i,treated} - Y_{i,comparison}$.

For equally weighted one-to-one pairs,

$$\hat{\tau}_{match} = mean(D_i).$$

This equals the difference in matched-arm means.

A causal interpretation still requires adequate adjustment and support.

Pairing supplies a comparison. Assumptions justify treating it causally.

R: compute the matched point estimate

```
md <- match.data(m1)
mu1 <- with(md, weighted.mean(re78[treat == 1],
                              weights[treat == 1]))
mu0 <- with(md, weighted.mean(re78[treat == 0],
                              weights[treat == 0]))
matched_difference <- mu1 - mu0
matched_difference
```

Keep the point estimate separate from the inference step.

Results depend on design and population

Analysis	Observed earnings contrast	Population represented
Crude full-sample	About −$635	Original observed groups
Initial nearest match	About +$894	All treated participants
0.05-caliper match	About +$1,433	113 retained treated participants

The change in estimates is not proof that the final design has removed confounding.

Uncertainty and remaining assumptions

Matched observations require an analysis that respects the matching structure.

Control reuse and unequal weights require additional care.

Standard resampling is not universally valid for nearest-neighbor estimators.

No matching method removes unmeasured confounding or absent support.

A credible report includes uncertainty, balance, retention, and causal limitations.

Matching: Summary

Matching and g-computation offer different ways to use a valid adjustment set.

Scores organize comparisons, while covariate diagnostics assess the resulting design.

Calipers can improve comparability while changing the represented population.

Outcome estimates become meaningful only after design and assumption checks.

Next: weighting lets us change contributions rather than selecting pairs.